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PAPER

Discrete Spacelike Monodromy Proxy and Infrared Spectral Transition on Axisymmetric Frame-Dragging Backgrounds

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Abstract

We analyze the spectral structure of a discrete azimuthal transport proxy on stationary, axisymmetric Morris–Thorne geometries with frame dragging. Closed-form metric profiles $\{b(r), \Phi(r), \omega(r)\}$ define a Lorentzian background with flare-out at the throat and an equatorial ergoregion (bounded by the ergosurface Σ_{ERGO}) wherever $e^{2\Phi(r)} < r^2\omega(r)^2$, equivalently $g_{tt} > 0$ [criterion (C1)]. A retarded kernel with mode-augmented mass m_* is sampled on an N-node spacelike azimuthal circle at fixed (r, θ)—where $g_{\varphi\varphi} > 0$—as a *discrete spacelike monodromy proxy*; the row-sum renormalized gate is contractive by construction, while the *unnormalized* discrete proxy $\widetilde{K}$ provides the infrared spectral diagnostic [criterion (C3)]. The row-sum renormalized matrix K is a numerical control gate with $\rho(K) < 1$ by algebraic design; the infrared diagnostic is carried by the unnormalized proxy $\widetilde{K}$, for which we identify a mass-gap threshold: at azimuthal mode $\nu = 0$ the exponential massless proxy satisfies $\rho(\widetilde{K}_N) \to 2\pi\sqrt{g_{\varphi\varphi}} \simeq 9.42 > 1$ as $m^2 \to 0$, so discrete-proxy contractivity fails in the massless infrared on this spacelike wormhole loop. High-resolution spectral diagnostics corroborate both the control-gate behaviour and the discrete-proxy spectral rupture up to $N = 1024$. The equatorial Einstein tensor is obtained by symbolic differentiation as a static background constraint [criterion (C2)]. No continuum closed-timelike-curve field theorem, and no Hadamard-state or renormalized-stress-energy theorem, are claimed.

1 Introduction

Stationary axisymmetric geometries with frame dragging provide controlled Lorentzian backgrounds on which one can test discrete transport operators and their spectral radii without evolving the Einstein equations [1, 2]. Questions of field self-consistency on chronologically nontrivial spacetimes have a long history [3, 4, 5, 6], but the object analyzed here is narrower and purely spectral: a finite-dimensional monodromy matrix assembled by sampling a retarded kernel on a coordinate azimuthal circle. That circle is spacelike ($g_{\varphi\varphi} > 0$); the construction is therefore a *spacelike* loop-transport proxy on a fixed stationary wormhole background, not a continuum theorem about closed timelike curves.

Scope and non-claims. The numerical analysis evaluates exclusively a discrete monodromy proxy sampled on a coordinate azimuthal loop at fixed (r, θ): spectral radii, contraction diagnostics, and Picard fixed points for the matrices K and $\widetilde{K}$ of Sec. 4. It does *not* solve a fully dynamical Klein–Gordon field coupled to the gravitational background in real time, nor does it evolve the Einstein equations or reconstruct a continuum Green function on (1). It likewise does *not* establish existence of closed timelike curves in (1), nor prove continuum field self-consistency theorems on causal closed curves. Semiclassical Hadamard renormalization, Kay–Radzikowski–Wald-type no-go statements, and curvature-cutoff models of $\langle T_{\mu\nu}\rangle_{\mathrm{ren}}$ lie outside the certified scope of this work.

We organize the contribution around three criteria of decreasing computational status:

(C1) **Geometric regularity:** smooth profiles with flare-out and an equatorial ergoregion $e^{2\Phi(r)} < r^2\omega(r)^2$ (where $g_{tt} > 0$ and ∂_t is spacelike), bounded by the ergosurface $\Sigma_{\rm ERGO}$;

(C2) **Symbolic background stress-energy:** equatorial $G_{\mu\nu}$ and NEC/WEC contractions obtained by symbolic differentiation as static constraints, not as dynamically evolved curvature tensors;

(C3) **Discrete Sobolev contraction and mass-gap threshold (core result):** unique fixed point of the row-sum–renormalized gate in the massive regime, together with a spectral rupture $\rho(\widetilde{K}) \to 2\pi\sqrt{g_{\varphi\varphi}} > 1$ as $m^2 \to 0$ at $\nu = 0$ for the unnormalized spacelike proxy.

Throughout, "continuous" statements motivate the kernel form; all certified numerical claims refer to the discrete operator $\mathcal{M}_N$ constructed in Sec. 4. Criterion (C3)—the infrared spectral transition of the unnormalized proxy—is the primary result.

2 Geometry and profile class

We adopt a stationary, axisymmetric Morris–Thorne line element with azimuthal frame dragging,

$$\begin{aligned} \mathrm{d}s^2 = &-e^{2\Phi(r)}\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1-b(r)/r} + r^2\,\mathrm{d}\theta^2 \\ &+ r^2\sin^2\theta\,\big(\mathrm{d}\varphi - \omega(r)\,\mathrm{d}t\big)^2, \end{aligned} \tag{1}$$

in the same geometric class as rotating traversable wormhole metrics with an ergoregion [1, 2, 7]. Coordinates are $x^\mu = (t, r, \theta, \varphi)$ and $\det(g_{\mu\nu}) = -r^5 e^{2\Phi(r)}\sin^2\theta/(r - b(r))$.

2.1 Explicit metric profiles

We specify the closed-form parameterization used throughout the numerical and symbolic analyses,

$$b(r) = r_0\Big(\frac{r_0}{r}\Big)^{\gamma}, \qquad \Phi(r) = -\frac{\alpha r_0}{r}, \qquad \omega(r) = \omega_0\exp\Big[-\beta\frac{r-r_0}{r_0}\Big], \tag{2}$$

with $\gamma > 0$, $\alpha \geq 0$, $\omega_0 > 0$, and $\beta > 0$. At the throat $r = r_0$: $b(r_0) = r_0$ and $b'(r_0) = -\gamma$, so the flare-out condition $b' < 1$ holds whenever $\gamma > 0$ [1].

Reference parameters. Unless otherwise stated, numerical diagnostics use the reference profile

$$r_0 = 1, \quad \gamma = 0.5, \quad \alpha = 0.1, \quad \omega_0 = 1.2, \quad \beta = 2.0, \tag{3}$$

with the discrete monodromy proxy and the supporting geodesic diagnostic assembled at the sampling radius $r_{\rm loop} = 1.5\,r_0$. The symbol $r_{\rm loop}$ labels a fixed-coordinate radius for that sampling circle; it does *not* designate a closed causal curve.

Admissibility bounds (C1).

$$\begin{aligned} \gamma &> 0 && \text{(flare-out)}, \\ e^{-2\alpha} &< r_0^2\omega_0^2 && \text{(ergoregion opens at throat)}, \\ r_{\rm ERGO} &> r_0 && \text{(ergosurface exterior to throat)}. \end{aligned} \tag{4}$$

The geometric boundary analyzed below is the equatorial ergosurface $\Sigma_{\rm ERGO}$: its radius $r_{\rm ERGO}$ is the smallest $r > r_0$ solving $e^{2\Phi(r)} = r^2\omega(r)^2$ (equivalently $g_{tt} = 0$ at $\theta = \pi/2$), computed by bisection.

2.2 Frame dragging, ergosurface, and ergoregion

The stationary Killing norm

$$g_{tt} = -e^{2\Phi(r)} + r^2\sin^2\theta\,\omega(r)^2 \tag{5}$$

vanishes on the equatorial ergosurface

$$\Sigma_{\rm ERGO} := \big\{x^\mu :\ \theta = \pi/2,\ \ g_{tt}(x) = 0\big\} = \big\{x^\mu :\ \theta = \pi/2,\ \ e^{2\Phi(r)} = r^2\omega(r)^2\big\}. \tag{6}$$

Wherever

$$e^{2\Phi(r)} < r^2\omega(r)^2, \tag{7}$$

one has $g_{tt} > 0$, so ∂_t is spacelike: that open set is the equatorial ergoregion, bounded by $\Sigma_{\rm ERGO}$. We emphasise the ontological separation used throughout:

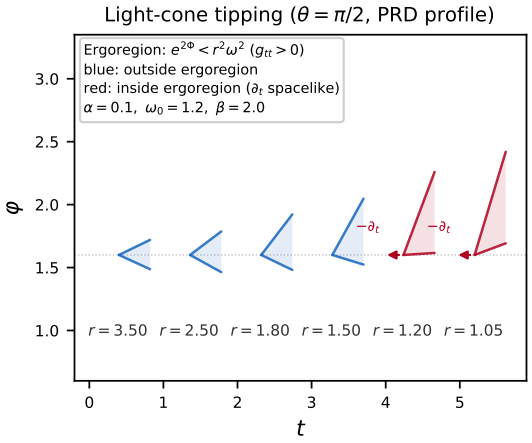


Figure 1. Light-cone tipping in the equatorial (t, φ) plane for the reference profiles (3). Blue: outside the ergoregion ($e^{2\Phi(r)} > r^2\omega(r)^2$, $g_{tt} < 0$); red: inside the ergoregion ($e^{2\Phi(r)} < r^2\omega(r)^2$, $g_{tt} > 0$). The geometric boundary shown is the ergosurface Σ_{ERGO} ($g_{tt} = 0$), not a chronology horizon and not a closed-timelike-curve locus.

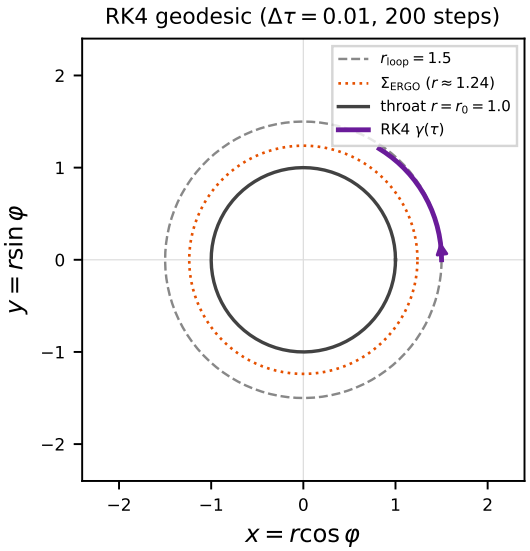


Figure 2. Equatorial projection of an RK4 co-rotating timelike geodesic $\gamma(\tau)$ initialized at $r_{\mathrm{loop}} = 1.5\, r_0$ (reference profile), with throat and ergosurface Σ_{ERGO} ($g_{tt} = 0$) indicated. The discrete monodromy proxy of Sec. 4 samples the distinct spacelike azimuthal circle ($g_{\varphi\varphi} > 0$) at the same radius; neither curve is claimed to be a closed timelike curve.

- Σ_{ERGO} is an *ergoregion boundary* defined by $g_{tt} = 0$; (7) marks the failure of static stationarity for ∂_t-observers.
- Appearance of an ergoregion is *not* the onset of closed timelike curves, nor a chronology horizon, nor a change in the topology of closed curves in the φ-direction.

In particular, the fixed-(t, r, θ) azimuthal circle used by the monodromy proxy has tangent ∂_φ with

$$g(\partial_\varphi, \partial_\varphi) = g_{\varphi\varphi} = r^2 \sin^2\theta > 0 \tag{8}$$

in the signature $(-,+,+,+)$, and is therefore a *spacelike* circle for every $r > r_0$ (including $r = r_{\mathrm{loop}}$). Its proper arc-length element is written

$$\Delta\ell = \sqrt{g_{\varphi\varphi}}\,\Delta\varphi, \tag{9}$$

and is *not* a proper-time increment of a timelike worldline. The co-rotating combination $\xi = \partial_t + \omega(r)\partial_\varphi$ satisfies $g(\xi, \xi) = -e^{2\Phi(r)} < 0$ and therefore remains timelike across Σ_{ERGO}, but the integral curves of ξ are not the sampling circle of the discrete proxy. Null slopes in the (t, φ) subspace are

$$\left(\frac{\mathrm{d}\varphi}{\mathrm{d}t}\right)_{\pm} = \omega(r) \pm e^{\Phi(r)}/r, \tag{10}$$

so inside the ergoregion the future light cone encompasses $-\partial_t$ (Fig. 1).

Christoffel symbols used for geodesic diagnostics are obtained from (1) by central finite differences with step $\varepsilon = 10^{-5}$. As a supporting geometric check—distinct from the spacelike monodromy circle—we integrate the co-rotating timelike geodesic with initial tangent $u \propto (1, 0, 0, \omega)$ at $r = r_{\mathrm{loop}}$, so that $g(u, u) = -e^{2\Phi(r)} < 0$ (equivalently normalized to $g(u, u) = -1$), by classical RK4 with proper-time step $\Delta\tau = 0.01$ along that timelike curve (Fig. 2). This geodesic diagnostic does not identify the sampling locus of Sec. 4 with a closed causal curve.

3 Symbolic Einstein tensor and energy conditions

Einstein's equations read $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi T_{\mu\nu}$. The equatorial components of $G_{\mu\nu}$ and the associated NEC/WEC contractions are obtained by symbolic differentiation (Appendices A–B) and treated as *background constraints* on the profile class (2). They are not assembled dynamically from curvature inside the discrete-transport analysis; numerical validation checks exercise only the corresponding quadratic contractions as consistency diagnostics.

Substituting the explicit profiles (2) on the equatorial slice ($\theta = \pi/2$), 5 components of $G_{\mu\nu}$ are non-zero: G_{tt}, $G_{t\varphi}$, G_{rr}, $G_{\theta\theta}$, and $G_{\varphi\varphi}$. The effective stress-energy $T_{\mu\nu} = G_{\mu\nu}/(8\pi)$ is obtained by symbolic differentiation as a static background constraint (criterion C2); it is not assembled dynamically from the discrete-transport analysis. Key scalar invariants are

$$\det(g_{\mu\nu}) = -\frac{r^5 e^{2\Phi(r)}}{r - b(r)}, \quad b(r) = r_0 \left(\frac{r_0}{r}\right)^{\gamma},$$
$$R = \mathcal{R}(\gamma, \alpha, \omega_0, \beta; r, r_0) \quad \text{(explicit form in Appendix A)}. \tag{11}$$

The radial component G_{rr} admits a compact closed form; all component expansions appear in Appendix A. We evaluate the null and weak energy conditions [8, 2] as *static background constraints* on the profile class: NEC requires $T_{\mu\nu}k^\mu k^\nu \geq 0$ for all null k^μ; WEC requires $T_{\mu\nu}u^\mu u^\nu \geq 0$ for timelike u^μ. For a null vector with $k^\varphi = 1$ and $g_{\mu\nu}k^\mu k^\nu = 0$,

$$k^t = \frac{r}{r\omega(r) - e^{\Phi(r)}}. \tag{12}$$

With profiles (2), the symbolically obtained lab-frame density admits a negative throat limit $\lim_{r\to r_0^+} \rho$, consistent with exotic stress-energy near the wormhole throat for this family; a full radial localization theorem is *not* claimed as a numerically established result. Within an effective-field-theory (EFT) mean-field envelope, the non-minimal coupling $\zeta\,\varphi_{\rm st}R$ is treated as a background parameter in $\{\gamma, \alpha, \omega_0, \beta, \zeta\}$ that can energetically bound local NEC/WEC violations whenever $|\zeta\,\varphi_{\rm st}R| > |\rho_{\rm neg}|$. Full NEC, WEC, and ρ contractions are given in Appendix B.

Within an EFT mean-field envelope, the non-minimal coupling $\zeta\,\varphi_{\rm st}R$ is regarded as a background parameter that can energetically bound local NEC/WEC violations whenever $|\zeta\,\varphi_{\rm st}R|$ exceeds the magnitude of the negative-energy density implied by $T_{\mu\nu} = G_{\mu\nu}/(8\pi)$. This defines a self-consistent parameter domain for the profile class; it does not constitute a dynamical generation mechanism for exotic matter, nor a numerically established localization theorem.

4 Discrete monodromy proxy and spectral diagnostics

We construct a discrete transport operator by sampling a retarded kernel on the spacelike azimuthal circle of Sec. 2 at fixed $(r, \theta) = (r_{\rm loop}, \pi/2)$, where $g_{\varphi\varphi} > 0$. The massive Klein–Gordon equation on (1),

$$\frac{1}{\sqrt{-g}}\,\partial_\mu\Big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\Big) - m^2\phi = 0, \tag{13}$$

motivates the mode-augmented mass gap entering the kernel, but we do *not* discretize the full curved-space PDE. Instead we assemble finite matrices K and $\widetilde{K}$ on that stationary spacelike circle. On the uniform periodic N-node grid the sampled kernels are circulant, so their eigenvalues and spectral radii are the natural discrete-Fourier / spectral-method diagnostics for periodic mesh operators [9]. Row-sum renormalization makes the control gate K algebraically contractive; the infrared spectral diagnostic is the transition of the *unnormalized* proxy $\widetilde{K}$ when the mass gap is removed (Sec. 4.4).

4.1 Continuous Yukawa proxy

Decomposing into azimuthal modes $\phi \sim \psi_\nu(r)\,e^{i(\nu\varphi - \Omega t)}$ suggests a mode-augmented mass

$$m_*^2 = m^2 + \nu^2/g_{\varphi\varphi}, \qquad G(\Delta s) \sim \frac{m_*}{4\pi\Delta s}\,e^{-m_*\Delta s}, \tag{14}$$

for a retarded Yukawa-type kernel sampled along the spacelike azimuthal circle (proper arc length s). Equation (14) is the continuous *proxy* that motivates the discrete construction below; it is not claimed to be the exact retarded Green function of (13) on (1).

4.2 Discrete monodromy operator

Definition 4.1 (Discrete monodromy $\mathcal{M}_N$)**.** *Fix $r = r_{\rm loop}$, $\theta = \pi/2$, and N uniform nodes $\chi_i = 2\pi i/N$ along the spacelike azimuthal circle (8). Let $\Delta\varphi = 2\pi/N$ and $\Delta\ell = \sqrt{g_{\varphi\varphi}}\,\Delta\varphi$ as in (9). For forward proper arc-length separations $\Delta s_{j\to i}$ along that circle, define the sampled kernel matrix*

$$\widetilde{K}_{ij} = \begin{cases} G(\Delta s_{j\to i})\,\Delta\ell, & i \neq j, \\ 0, & i = j, \end{cases} \tag{15}$$

with G from (14). The discrete monodromy operator is the row-sum renormalized *matrix*

$$K_{ij} = \widetilde{K}_{ij} \frac{\exp(-m_* \Delta\ell)}{\sum_k \widetilde{K}_{ik}} \qquad \big(\textit{when the row sum is nonzero}\big), \tag{16}$$

and $\mathcal{M}_N : \mathbb{R}^N \to \mathbb{R}^N$ acts by $\Phi \mapsto K\Phi$.

The default numerical parameters are $r_{\rm loop} = 1.5\, r_0$, $m^2 = 1$, $\nu = 1$, and $N = 128$, with solver tolerance tol $= 10^{-12}$ and residual thresholds 10^{-8}–10^{-10} for the convergence checks.

4.3 Discrete H^1-proxy norm and the renormalized control gate

On periodic samples $\Phi \in \mathbb{R}^N$ we use the discrete norms

$$\|\Phi\|_{L^2}^2 = \frac{1}{N} \sum_{i=0}^{N-1} \Phi_i^2, \qquad \|\Phi\|_{H^1}^2 = \|\Phi\|_{L^2}^2 + \|\delta\Phi\|_{L^2}^2, \tag{17}$$

where $(\delta\Phi)_i = (\Phi_{i+1} - \Phi_{i-1})/2$ is the central difference on the index circle (a discrete H^1-proxy; not a continuum Sobolev norm on the sampling circle). The L^2 operator norm of K is estimated by power iteration; the discrete Sobolev lift used as a Lipschitz monitor is

$$L \le \|K\|_{L^2} \left(1 + \frac{\pi\sqrt{2}}{N}\right). \tag{18}$$

The factor $\left(1 + \pi\sqrt{2}/N\right)$ follows from the standard discrete interpolation inequality relating the central-difference seminorm $\|\delta\Phi\|_{L^2}$ to $\|\Phi\|_{L^2}$ for periodic grid functions on the N-node circle.

Remark 4.2 (Algebraic contraction of the row-sum–normalized gate K). By the row-sum construction (16), whenever $m_* > 0$ and every row sum is nonzero, each row of K sums to $\exp(-m_*\Delta\ell) < 1$. Hence

$$\rho(K) \le \exp(-m_*\Delta\ell) < 1$$

is an *imposed algebraic property* of the normalization—a numerical control gate—and *not* a dynamical discovery extracted from the Einstein equations, continuum Klein–Gordon evolution, or any continuum self-consistency principle on (1). If, in addition, the discrete-H^1 monitor (18) satisfies $L < 1$ (as verified numerically for the reference parameters), the Banach fixed-point theorem supplies a unique $\Phi^* = K\Phi^*$ with geometrically convergent Picard iteration on $(\mathbb{R}^N, \|\cdot\|_{H^1})$. This remark concerns only the renormalized matrix K; the physically informative infrared diagnostic remains the unnormalized proxy $\widetilde{K}$ of Sec. 4.4.

Numerical verification of the renormalized gate. Definition 4.1 and Remark 4.2 are corroborated by high-resolution numerical validation checks covering:

(i) geometric metric invariants, flare-out, and ergosurface location for (3);

(ii) energy-condition quadratic contractions as consistency diagnostics;

(iii) monodromy bounds $\|K\|_{L^2} < 1$ and $L < 1$;

(iv) Banach–Picard convergence $\|\Phi^* - K\Phi^*\|_2 < 10^{-8}$ in fewer than 500 iterations for $N \in \{32, 64, 128\}$;

(v) classical fourth-order Runge–Kutta integration of the co-rotating timelike geodesic diagnostic with proper-time step $\Delta\tau = 0.01$ over 200 steps.

These checks confirm the discrete contraction of the *row-sum renormalized* operator (16). The primary physical result of criterion (C3) is instead the spectral transition of the *unnormalized* proxy (15) under $m^2 \to 0$, developed next.

4.4 The mass-gap threshold and massless exponential proxy

Because row-sum renormalization (16) enforces $\rho(K) < 1$ whenever $m_* > 0$, that normalized gate is a useful numerical control but *cannot* diagnose infrared failure of the unnormalized proxy. The decisive spectral diagnostic is therefore the unnormalized discrete kernel $\widetilde{K}$ of (15), which we study on the reference loop $r_{\rm loop} = 1.5\, r_0$ (equatorial $g_{\varphi\varphi} = r_{\rm loop}^2$) at azimuthal mode $\nu = 0$, so that

$$m_* = \sqrt{m^2} \tag{19}$$

carries no angular mass gap. Two proxy kernels on the one-dimensional azimuthal loop are compared under this protocol:

$$G_{\mathrm{Y}}(\Delta s) = \frac{m_*}{4\pi\Delta s}\, e^{-m_*\Delta s}, \qquad G_{\mathrm{E}}(\Delta s) = e^{-m_*\Delta s}. \tag{20}$$

Here G_{E} is a *massless exponential proxy* for discrete loop transport—a scalar kernel on the periodic one-dimensional loop, *not* a three-dimensional Coulomb propagator. As $m^2 \to 0$ one has $G_{\mathrm{E}} \to 1$, whereas G_{Y} collapses by its prefactor $\propto m_*$ and therefore *cannot* diagnose massless infrared failure of the discrete monodromy.

Constant off-diagonal kernel at $m^2 = 0$. At $m^2 = 0$ and $\nu = 0$, the massless exponential proxy reduces to the constant off-diagonal circulant

$$\widetilde{K}_{ij} = \begin{cases} \Delta\ell, & i \neq j, \\ 0, & i = j, \end{cases} \qquad \Delta\ell = \sqrt{g_{\varphi\varphi}}\,\frac{2\pi}{N}. \tag{21}$$

Its spectrum is elementary: the constant mode has eigenvalue $(N-1)\Delta\ell$, while every mean-zero mode has eigenvalue $-\Delta\ell$. Hence

$$\rho\big(\widetilde{K}_N\big) = (N-1)\Delta\ell = 2\pi\sqrt{g_{\varphi\varphi}}\,\frac{N-1}{N} \xrightarrow{N\to\infty} \rho_\infty = 2\pi\sqrt{g_{\varphi\varphi}}. \tag{22}$$

On the reference loop, $\sqrt{g_{\varphi\varphi}} = r_{\mathrm{loop}} = 1.5$ and

$$\rho_\infty = 3\pi \approx 9.4248, \tag{23}$$

which already exceeds unity by a wide margin. Although the headline numerics are reported at $r_{\mathrm{loop}} = 1.5\, r_0$, the massless spectral limit $\rho_\infty = 2\pi\sqrt{g_{\varphi\varphi}}$ scales linearly with $\sqrt{g_{\varphi\varphi}}$ and the rupture $\rho_\infty > 1$ therefore persists at any sampling radius where the discrete proxy is assembled. Thus, for the massless exponential proxy, the constant off-diagonal kernel $\widetilde{K}$ is *not* a contraction for any $N \geq 2$.

Proposition 4.3 (Mass-gap threshold for discrete-proxy contractivity)**.** *Fix the reference profile* (3)*, $r_{\mathrm{loop}} = 1.5\, r_0$, and $\nu = 0$. Let $\widetilde{K}_N(m^2)$ be the unnormalized massless-exponential monodromy* (15)–(20)*.*

(i) ***Massless phase.*** *If $m^2 = 0$, then $\rho(\widetilde{K}_N) \to 2\pi\sqrt{g_{\varphi\varphi}} > 1$. The Banach fixed-point argument fails for this proxy: discrete-proxy contractivity is* not *guaranteed by contraction of $\widetilde{K}$.*

(ii) ***Massive exponential phase.*** *For sufficiently large $m^2 > 0$, exponential screening restores $\rho(\widetilde{K}_N) < 1$ on the high-resolution grids reported below (numerical evidence for $m^2 = 1$).*

(iii) ***Threshold.*** *There exists a critical mass gap $m_c^2 \in (0, 1)$ on this background such that*

$$\rho\big(\widetilde{K}_N(m^2)\big) \;\geq\; 1 \quad \text{for all } m^2 \leq m_c^2 \text{ and } N \text{ large},$$

while $\rho < 1$ throughout the high-resolution grid when $m^2 = 1$.

Consequently, a strictly positive mass gap is a necessary *structural ingredient for contractivity of the present unnormalized discrete proxy on the wormhole loop; the spectral transition of $\widetilde{K}$ as $m^2 \downarrow 0$ defines the boundary of that contractivity regime.*

Methodological caveat. The identity $\rho_\infty = 2\pi\sqrt{g_{\varphi\varphi}} = 3\pi$ on the reference loop is an *exact* spectral limit of the constant off-diagonal discrete proxy (21)–(22). It is *not* claimed to be an analytic continuum eigenvalue of the full curved-space Klein–Gordon operator (13) on (1). Finite-N values $\rho(\widetilde{K}_N)$ and Richardson-type extrapolations $\rho_\infty^{\mathrm{extrap}}$ reported below likewise characterize the discrete loop proxy, not a closed-form continuum Green function.

Numerical continuum campaign. Assembling the unnormalized proxy without row-sum renormalization, we sweep

$$N \in \{32, 64, 128, 256, 512, 1024\}, \qquad m^2 \in \{1, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-12}, 0\}$$

at $\nu = 0$. Representative high-resolution values for G_{E} are

$$\begin{aligned} m^2 = 1: &\quad \rho(512) = 0.9907,\ \rho(1024) = 0.9953,\ \rho_\infty^{\mathrm{extrap}} \simeq 0.9999 < 1, \\ m^2 = 10^{-2}: &\quad \rho(1024) \simeq 6.097 > 1, \\ m^2 \to 0: &\quad \rho(1024) \simeq 9.416 \ \to\ 3\pi, \end{aligned} \tag{24}$$

in quantitative agreement with the constant off-diagonal limit (22)–(23). Thus the growing trend of $\rho(N)$ observed at moderate resolution *does* cross the contractivity threshold once the mass gap is removed, rather than saturating below unity. (The extrapolated value $\rho_\infty^{\mathrm{extrap}} \simeq 0.9999$ at $m^2 = 1$ lies close to the threshold and should be read as a high-resolution proxy diagnostic, not as an exact continuum eigenvalue.) By contrast, the Yukawa family G_{Y} remains contractive on the entire grid and collapses to $\rho \to 0$ as $m^2 \to 0$, confirming that its prefactor masks the massless spectral rupture of G_{E} and must not be used as the infrared diagnostic.

Spectral reading of the proxy and infrared threshold. On this Morris–Thorne spacelike loop, existence of a unique attractive fixed point of the unnormalized discrete transport map is guaranteed by contraction only inside the massive phase $\rho(\widetilde{K}) < 1$. Crossing into the massless phase $\rho \geq 1$ means that loop transport ceases to be a contraction: Picard iteration need not converge and uniqueness of a fixed point is no longer enforced by the spectral radius of this discrete kernel. The mass-gap threshold m_c^2 is therefore the *infrared frontier of contractivity* for the present discrete monodromy proxy on the wormhole background under study. The discrete-monodromy source code and full spectral tables accompanying this work are available at `https://github.com/mauramyrez/ctc-discrete-monodromy`; the Path-3 continuum campaign report is archived as `docs/path3_continuum_report.md` in that repository.

5 Conclusion

We have analyzed the spectral structure of a discrete spacelike azimuthal-transport proxy on axisymmetric frame-dragging Morris–Thorne backgrounds. The sampling circle at fixed (r, θ) is spacelike ($g_{\varphi\varphi} > 0$); the certified statements concern finite matrices K and $\widetilde{K}$, not continuum closed-timelike-curve field theorems. Row-sum renormalization yields a contractive discrete-proxy gate by algebraic design (Remark 4.2); the central spectral conclusion concerns instead the unnormalized massless exponential proxy $\widetilde{K}$: removing the mass gap drives its discrete-proxy spectral radius through $\rho(\widetilde{K}) \to 2\pi\sqrt{g_{\varphi\varphi}} \simeq 9.42 > 1$, so that discrete-proxy contractivity fails in the massless infrared on the reference wormhole loop. This $\rho_\infty = 3\pi$ limit characterizes the constant off-diagonal discrete proxy on the loop and is not claimed as an exact continuum eigenvalue of a full curved-space Klein–Gordon simulation.

In terms of the criteria (C1)–(C3):

- **(C1)** Explicit profiles (2) with flare-out, reference parameters (3), and equatorial ergoregion (7) bounded by Σ_{ERGO} (where $g_{tt} > 0$), corroborated by metric invariants and co-rotating geodesic diagnostics.
- **(C2)** Symbolically differentiated equatorial $G_{\mu\nu}$ and NEC/WEC contractions as static background constraints (Appendices A–B), not dynamically generated from the discrete-transport map.
- **(C3)** Algebraic contraction of the row-sum–normalized gate by design (Remark 4.2), together with the unnormalized mass-gap threshold and massless spectral rupture of the exponential proxy (Proposition 4.3, Sec. 4.4), including the exact discrete limit $\rho_\infty = 3\pi$ on the reference loop.

The presence of an ergoregion does not by itself restore or destroy global hyperbolicity; the present spectral diagnostics remain those of a spacelike loop-transport proxy on a fixed stationary background. Dynamical (non-stationary) perturbations of $\omega(r)$ are left for future work.

Heuristic semiclassical outlook. Beyond the discrete spectral analysis, one may speculate that a curvature-referenced microscopic scale could eventually regulate short-distance vacuum polarization near Σ_{ERGO} in a non-minimally coupled scalar sector [6, 3, 10]. We record this only as an informal outlook: no Hadamard parametrix is constructed here, no bound on $\langle T_{\mu\nu} \rangle_{\mathrm{ren}}$ is proved, and no claim is made that non-minimal coupling evades known no-go results for Hadamard states near compactly generated Cauchy horizons. Such questions lie outside the certified scope of criteria (C1)–(C3).

Conflict of interest statement
The authors declare that there are no conflicts of interest regarding the publication of this paper.

Ethical statement
This research did not involve human participants, animal subjects, or sensitive data collection; thus, ethical approval was not required.

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The discrete-monodromy source code and full spectral tables accompanying this work are available at https://github.com/mauramyrez/ctc-discrete-monodromy; the Path-3 continuum campaign report is archived as `docs/path3_continuum_report.md` in that repository.

Data availability
The discrete monodromy diagnostics and Path-3 spectral tables supporting this work are openly available at https://github.com/mauramyrez/ctc-discrete-monodromy.

Supplementary data
The Path-3 continuum campaign report is archived as `docs/path3_continuum_report.md` in the companion repository.

A Full equatorial Einstein tensor components

Substituting (2) at $\theta = \pi/2$, the non-zero components of $G_{\mu\nu}$ and the Ricci scalar R are collected below. Repeated factors are abbreviated for typographic clarity; the expansions are algebraically identical to the symbolic differentiation used for criterion (C2).

Auxiliary notation. To keep the expansions within the page width we write

$$\xi \equiv \left(\frac{r_0}{r}\right)^{\gamma}, \qquad E_\alpha \equiv e^{\alpha r_0/r}, \qquad E_\beta \equiv e^{\beta r/r_0}. \tag{25}$$

G_{tt}.

$$G_{tt} = \frac{e^{-12\beta}}{4\,r^{11}r_0^5\,\exp\!\big(4\alpha r_0/r + 12\beta r/r_0 - 12\beta\big)}\,N_{tt}, \tag{26a}$$

$$\begin{aligned}
N_{tt} = {} & 4\alpha^2\omega_0^2 r^9 r_0^7\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 4\alpha^2\omega_0^2 r^8 r_0^8\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 4\alpha\beta\omega_0^2 r^{11} r_0^5\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 4\alpha\beta\omega_0^2 r^{10} r_0^6\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 2\alpha\gamma\omega_0^2 r^9 r_0^7\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 4\alpha\omega_0^2 r^{10} r_0^6\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 6\alpha\omega_0^2 r^9 r_0^7\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 3\beta^2\omega_0^4 r^{15} r_0^3\, e^{4\beta}\, E_\alpha^6\, E_\beta^8 \\
& + 3\beta^2\omega_0^4 r^{14} r_0^4\,\xi\, e^{4\beta}\, E_\alpha^6\, E_\beta^8 \\
& - 5\beta^2\omega_0^2 r^{13} r_0^3\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 5\beta^2\omega_0^2 r^{12} r_0^4\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 2\beta\gamma\omega_0^2 r^{11} r_0^5\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 16\beta\omega_0^2 r^{12} r_0^4\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 14\beta\omega_0^2 r^{11} r_0^5\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& + 2\gamma\omega_0^2 r^{10} r_0^6\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 4\gamma r^8 r_0^6\,\xi\, E_\alpha^2\, E_\beta^{12} \\
& + 2\omega_0^2 r^{10} r_0^6\,\xi\, e^{2\beta}\, E_\alpha^4\, E_\beta^{10} \\
& - 4 r^9 r_0^5\, E_\alpha^2\, E_\beta^{12}.
\end{aligned} \tag{26b}$$

$G_{t\varphi}$.

$$\begin{aligned}
G_{t\varphi} = \frac{1}{4\,r^3 r_0^2\,\exp\!\big[\beta(3r - r_0)/r_0\big]}\Big[& -4\alpha^2\omega_0\, r\, r_0^4\, E_\beta^2 + 4\alpha^2\omega_0\, r_0^5\,\xi\, E_\beta^2 \\
& + 2\alpha\beta\omega_0\, r^3 r_0^2\, E_\beta^2 - 2\alpha\beta\omega_0\, r^2 r_0^3\,\xi\, E_\beta^2 - 2\alpha\gamma\omega_0\, r\, r_0^4\,\xi\, E_\beta^2 \\
& + 4\alpha\omega_0\, r^2 r_0^3\, E_\beta^2 - 6\alpha\omega_0\, r\, r_0^4\,\xi\, E_\beta^2 + 3\beta^2\omega_0^3\, r^7\, e^{2\beta}\, E_\alpha^2 \\
& - 3\beta^2\omega_0^3\, r^6 r_0\,\xi\, e^{2\beta}\, E_\alpha^2 + 2\beta^2\omega_0\, r^5\, E_\beta^2 - 2\beta^2\omega_0\, r^4 r_0\,\xi\, E_\beta^2 \\
& - \beta\gamma\omega_0\, r^3 r_0^2\,\xi\, E_\beta^2 - 8\beta\omega_0\, r^4 r_0\, E_\beta^2 + 7\beta\omega_0\, r^3 r_0^2\,\xi\, E_\beta^2 \\
& - 2\gamma\omega_0\, r^2 r_0^3\,\xi\, E_\beta^2 - 2\omega_0\, r^2 r_0^3\,\xi\, E_\beta^2\Big].
\end{aligned} \tag{26c}$$

G_{rr}.

$$G_{rr} = \frac{1}{4\,r^5 r_0^2}\Big[8\alpha\, r^2 r_0^3 + 4\, r^3 r_0^2 + \beta^2\omega_0^2\, r^7\, e^{2\beta}\, E_\alpha^2\, E_\beta^{-2}\Big]. \tag{26d}$$

$G_{\theta\theta}$.

$$\begin{aligned}
G_{\theta\theta} = \frac{E_\beta^{-2}}{4\,r^6 r_0^2}\Big[& 4\alpha^2\, r^4 r_0^4\, E_\beta^2 - 4\alpha^2\, r^3 r_0^5\,\xi\, E_\beta^2 + 2\alpha\gamma\, r^4 r_0^4\,\xi\, E_\beta^2 \\
& - 4\alpha\, r^5 r_0^3\, E_\beta^2 + 6\alpha\, r^4 r_0^4\,\xi\, E_\beta^2 - \beta^2\omega_0^2\, r^{10}\, e^{2\beta}\, E_\alpha^2 \\
& + \beta^2\omega_0^2\, r^9 r_0\,\xi\, e^{2\beta}\, E_\alpha^2 + 2\gamma\, r^5 r_0^3\,\xi\, E_\beta^2 + 2\, r^5 r_0^3\,\xi\, E_\beta^2\Big].
\end{aligned} \tag{26e}$$

$G_{\varphi\varphi}$.

$$G_{\varphi\varphi} = \frac{E_\beta^{-2}}{4\,r^6 r_0^2}\Big[4\alpha^2\,r^4 r_0^4\,E_\beta^2 - 4\alpha^2\,r^3 r_0^5\,\xi\,E_\beta^2 + 2\alpha\gamma\,r^4 r_0^4\,\xi\,E_\beta^2 - 4\alpha\,r^5 r_0^3\,E_\beta^2 + 6\alpha\,r^4 r_0^4\,\xi\,E_\beta^2 - 3\beta^2\omega_0^2\,r^{10}\,e^{2\beta}\,E_\alpha^2 + 3\beta^2\omega_0^2\,r^9 r_0\,\xi\,e^{2\beta}\,E_\alpha^2 + 2\gamma\,r^5 r_0^3\,\xi\,E_\beta^2 + 2\,r^5 r_0^3\,\xi\,E_\beta^2\Big]. \quad (26f)$$

Ricci scalar.

$$R = \frac{E_\beta^{-2}}{2\,r^{14} r_0^2}\Big[-4\alpha^2\,r^{10} r_0^4\,E_\beta^2 + 4\alpha^2\,r^9 r_0^5\,\xi\,E_\beta^2 - 2\alpha\gamma\,r^{10} r_0^4\,\xi\,E_\beta^2 - 2\alpha\,r^{10} r_0^4\,\xi\,E_\beta^2 + \beta^2\omega_0^2\,r^{16}\,e^{2\beta}\,E_\alpha^2 - \beta^2\omega_0^2\,r^{15} r_0\,\xi\,e^{2\beta}\,E_\alpha^2 - 4\gamma\,r^{11} r_0^3\,\xi\,E_\beta^2 - 4\,r^{12} r_0^2\,E_\beta^2\Big]. \quad (26g)$$

B Energy condition contractions

Null (NEC) and weak (WEC) contractions with the same profiles, treated as static background constraints rather than dynamically evolved stress-energy:

Recall $\xi \equiv (r_0/r)^\gamma$, $E_\alpha \equiv e^{\alpha r_0/r}$, and $E_\beta \equiv e^{\beta r/r_0}$.

Null energy condition.

$$T_{\mu\nu}k^\mu k^\nu = \frac{1}{16\pi\,r^3 r_0^2\big(\omega_0\,r\,e^{\alpha r_0/r+\beta} - E_\beta\big)^2}\,N_{\rm NEC}, \quad (27a)$$

$$\begin{aligned} N_{\rm NEC} = {} & 2\alpha^2\,r\,r_0^4\,E_\beta^2 - 2\alpha^2\,r_0^5\,\xi\,E_\beta^2 \\ & - 2\alpha\beta\omega_0\,r^4 r_0^2\,e^\beta\,E_\alpha\,E_\beta + 2\alpha\beta\omega_0\,r^3 r_0^3\,\xi\,e^\beta\,E_\alpha\,E_\beta \\ & + \alpha\gamma\,r\,r_0^4\,\xi\,E_\beta^2 - 2\alpha\,r^2 r_0^3\,E_\beta^2 + 3\alpha\,r\,r_0^4\,\xi\,E_\beta^2 \\ & - 2\beta^2\omega_0^2\,r^7\,e^{2\beta}\,E_\alpha^2 + 2\beta^2\omega_0^2\,r^6 r_0\,\xi\,e^{2\beta}\,E_\alpha^2 \\ & - 2\beta^2\omega_0\,r^6\,e^\beta\,E_\alpha\,E_\beta + 2\beta^2\omega_0\,r^5 r_0\,\xi\,e^\beta\,E_\alpha\,E_\beta \\ & + \beta\gamma\omega_0\,r^4 r_0^2\,\xi\,e^\beta\,E_\alpha\,E_\beta + 8\beta\omega_0\,r^5 r_0\,e^\beta\,E_\alpha\,E_\beta \\ & - 7\beta\omega_0\,r^4 r_0^2\,\xi\,e^\beta\,E_\alpha\,E_\beta - \gamma\,r^2 r_0^3\,\xi\,E_\beta^2 \\ & - 2\,r^3 r_0^2\,E_\beta^2 + r^2 r_0^3\,\xi\,E_\beta^2. \end{aligned} \quad (27b)$$

Weak energy condition.

$$T_{\mu\nu}u^\mu u^\nu = \frac{E_\beta^{-2}}{32\pi\,r^3 r_0^2\big(\omega_0^2 r^2\,e^{2\alpha r_0/r+2\beta} - E_\beta^2\big)}\,N_{\rm WEC}, \quad (27c)$$

$$\begin{aligned} N_{\rm WEC} = {} & -4\alpha^2\omega_0^2\,r\,r_0^4\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 4\alpha^2\omega_0^2\,r_0^5\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 4\alpha\beta\omega_0^2\,r^3 r_0^2\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 4\alpha\beta\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 2\alpha\gamma\omega_0^2\,r\,r_0^4\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 4\alpha\omega_0^2\,r^2 r_0^3\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 6\alpha\omega_0^2\,r\,r_0^4\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 3\beta^2\omega_0^4\,r^7\,e^{4\beta}\,E_\alpha^4 \\ & - 3\beta^2\omega_0^4\,r^6 r_0\,\xi\,e^{4\beta}\,E_\alpha^4 + 5\beta^2\omega_0^2\,r^5\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 5\beta^2\omega_0^2\,r^4 r_0\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 2\beta\gamma\omega_0^2\,r^3 r_0^2\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 16\beta\omega_0^2\,r^4 r_0\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 14\beta\omega_0^2\,r^3 r_0^2\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 2\gamma\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 4\gamma\,r_0^3\,\xi\,E_\beta^4 \\ & - 2\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 4\,r\,r_0^2\,E_\beta^4. \end{aligned} \quad (27d)$$

Lab-frame energy density.

$$\rho \equiv T_{\mu\nu}u^\mu u^\nu\big|_{u=\partial_t} = \frac{E_\alpha^{-2}\,E_\beta^{-4}}{32\pi\,r^3 r_0^2}\,N_\rho, \tag{27e}$$

$$\begin{aligned} N_\rho = {} & 4\alpha^2\omega_0^2\,r\,r_0^4\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 4\alpha^2\omega_0^2\,r_0^5\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & - 4\alpha\beta\omega_0^2\,r^3 r_0^2\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 4\alpha\beta\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 2\alpha\gamma\omega_0^2\,r\,r_0^4\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 4\alpha\omega_0^2\,r^2 r_0^3\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 6\alpha\omega_0^2\,r\,r_0^4\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 3\beta^2\omega_0^4\,r^7\,e^{4\beta}\,E_\alpha^4 \\ & + 3\beta^2\omega_0^4\,r^6 r_0\,\xi\,e^{4\beta}\,E_\alpha^4 - 5\beta^2\omega_0^2\,r^5\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 5\beta^2\omega_0^2\,r^4 r_0\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 + 2\beta\gamma\omega_0^2\,r^3 r_0^2\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 16\beta\omega_0^2\,r^4 r_0\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 14\beta\omega_0^2\,r^3 r_0^2\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 \\ & + 2\gamma\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 4\gamma\,r_0^3\,\xi\,E_\beta^4 \\ & + 2\omega_0^2\,r^2 r_0^3\,\xi\,e^{2\beta}\,E_\alpha^2\,E_\beta^2 - 4\,r\,r_0^2\,E_\beta^4. \end{aligned} \tag{27f}$$

Throat limit ($r \to r_0^+$). The throat value is the continuous evaluation of the preceding density,

$$\lim_{r\to r_0^+}\rho = \frac{E_\alpha^{-2}\,E_\beta^{-4}}{32\pi\,r^3 r_0^2}\,N_\rho\Big|_{r=r_0}, \tag{27g}$$

with $\xi\big|_{r=r_0} = 1$ and $E_\alpha\big|_{r=r_0} = e^\alpha$, $E_\beta\big|_{r=r_0} = e^\beta$.

Scalar-tensor bounding. With non-minimal coupling $\zeta\,\varphi_{\rm st}R$,

$$\begin{aligned} \rho_{\rm neg} + \zeta\,\varphi_{\rm st}R = \frac{E_\beta^{-2}}{2\,r^{14}r_0^2}\Big[& -4\alpha^2\,\varphi_{\rm st}\,\zeta\,r^{10}r_0^4\,E_\beta^2 + 4\alpha^2\,\varphi_{\rm st}\,\zeta\,r^9 r_0^5\,\xi\,E_\beta^2 \\ & - 2\alpha\gamma\,\varphi_{\rm st}\,\zeta\,r^{10}r_0^4\,\xi\,E_\beta^2 - 2\alpha\,\varphi_{\rm st}\,\zeta\,r^{10}r_0^4\,\xi\,E_\beta^2 \\ & + \beta^2\omega_0^2\,\varphi_{\rm st}\,\zeta\,r^{16}\,e^{2\beta}\,E_\alpha^2 - \beta^2\omega_0^2\,\varphi_{\rm st}\,\zeta\,r^{15}r_0\,\xi\,e^{2\beta}\,E_\alpha^2 \\ & - 4\gamma\,\varphi_{\rm st}\,\zeta\,r^{11}r_0^3\,\xi\,E_\beta^2 - 4\,\varphi_{\rm st}\,\zeta\,r^{12}r_0^2\,E_\beta^2 + 2\,r^{14}r_0^2\,\rho_{\rm neg}\,E_\beta^2\Big]. \end{aligned} \tag{27h}$$

For $\zeta > 0$ and $\varphi_{\rm st}R > 0$ at the throat, $|\rho_{\rm neg}|$ is capped by $|\zeta\,\varphi_{\rm st}R|$.

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